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Business Information Server Software Overview

Business Information Server software is a product that lets you design and control your own database applications, without requiring knowledge of programming. For example, you can design a database containing information such as employee name, employee number, telephone, and address. You store this information in a portion of the database called a report. You can perform functions on the report, such as searching, sorting, and calculating.

When you become more familiar with Business Information Server software, you can automate repeated use of functions using scripts.

This topic introduces the Business Information Server software, including the following topics:

- Script Overview
- System Administration
- Database Structure
- Cabinet Overview
- Drawer Overview
- Report Overview
- Business Information Server Graphical Environment

Script Overview

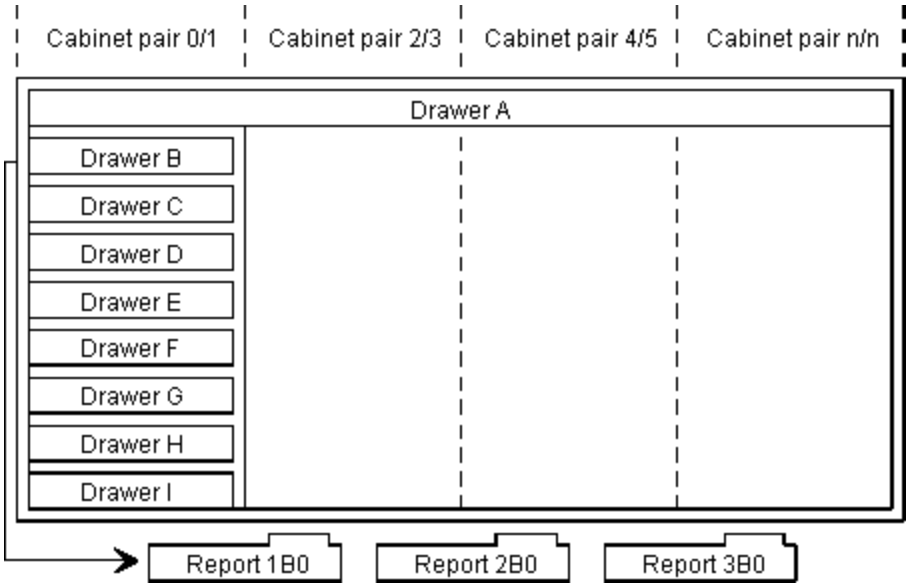
A script is a collection of functions that you execute together. For example, suppose you frequently find subtotals of a certain field in a report and then search the entire database for a given character string. You can automate these steps by typing statements in a special report called a run control report. You then execute this series of statements, called a script, by entering the script name. The script must be registered by the system administrator.

System Administration

The Business Information Server system administrator is the person who controls the development and use of the Business Information Server database and is responsible for its security. You must talk to the administrator, for example, to obtain a sign-on and to register scripts.

Database Structure

The method of organizing information within the database is similar to that of a traditional office. Think of your database as a large storage room. Within this room are filing cabinets containing drawers. These drawers, in turn, contain information in the form of reports. The database adapts this structure to an electronic database; even the terminology remains the same. You access cabinets, drawers, and reports to recall, review, and revise the information in your Business Information Server database. The following figure gives a visual representation of the database structure.



Cabinet Overview

You use sections of the database called cabinets (so named because they can be compared to filing cabinets) to store important information on a specific aspect of your organization or operation. For example, one cabinet might contain information on the accounting department for your organization, a second cabinet might be dedicated to the sales department, and a third to the shipping department.

Cabinets are numbered in pairs so that each cabinet has an even number and an odd number, starting with cabinet pair 0/1.

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Each cabinet in the pair holds exactly the same data, but you can update information in the even-numbered cabinet, and only read the information in the odd-numbered cabinet. This is one way the Business Information Server enables you to secure your data, letting some users update while other users can only look at the data.

The number of cabinet pairs varies from site to site. Default = 401 cabinet pairs. Maximum = 2048 cabinet pairs: cabinets 0 through 4095.

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The system administrator sets up security groups that allow you read access and write access to cabinets and drawers. For more information, see Security Features.

The system contains 2001 cabinet pairs: cabinets 0 through 4001.

Ü Ü Windows Linux/UNIX OS

Drawer Overview

Just as a filing cabinet contains drawers, each Business Information Server software cabinet contains eight drawers. Each drawer is a subcategory for the specific topic to which that cabinet is dedicated. For example, the cabinet for a sales department might contain one drawer for each salesperson.

Drawers are named by letter, A through I. Drawers can also have a substitute name. For information, see the *Command Reference.* Drawer A is the freeform drawer, which is like a blank sheet of paper that you can use for any sort of data or communication. Drawer A is shared by all cabinets, but it is not displayed in the drawer table of contents. It contains general information, which is not protected.

Windows Linux/UNIX OS Þ Þ

Read and write access to drawer A is restricted by a security group. If you need to gain access to drawer A, see your supervisor or system administrator.

Ü Ü Windows Linux/UNIX OS

Drawers B through I are your databases, laid out in either line and column format, or freeform, like drawer A.

One or more users plan the drawers and the administrator generates them.

Report Overview

You can use each drawer to store a large number of reports containing detailed information, typically on a line-by-line basis (for example, all sales closed during a specific month for a particular salesperson).

To identify reports, specify one of the following items:

- A report name (such as Monthly)
- A report number and drawer letter (such as 2B)
- A report number, drawer letter, and cabinet number (such as 2B0)

For information on how to identify your reports with a substitute name, see the *Command Reference.*

Reports can contain a limited number of lines. Use the LIMITS command to display the line limit. For the most efficient use of system resources, it is recommended that you have multiple reports with a reasonable number of lines, versus having one large report.

For details on the structure of Business Information Server reports, see Managing Reports.

Business Information Server Graphical Environment

The Business Information Server Graphical Environment provides a practical point-and-click interface to many common data access, ad-hoc reporting, and user administration tasks. This environment brings a graphical interface to the character-based manual functions. It consists of two main display screens (shells):

- The Active Screen shell, the equivalent of the active screen, lets you point and click to access existing reports in the database or to create new reports.
- The Report Display shell displays reports and results. It contains point-and-click tools to help you process, update, and analyze report data.

With the Business Information Server Graphical Environment, users and administrators have a choice between the traditional, character-based interface and a point-and-click interface. It is easy to switch between the two interfaces. You can either point to the Native Command edit box and enter a command string or click the Exit to Native Environment toolbar icon to display the active screen. The availability of the traditional character-based interface allows experienced users to perform more complex tasks or to use the advanced data analysis features of the Business Information Server software in a familiar manner.

The Graphical Environment provides additional features that simplify the work of both end-users and administrators.

Intuitive Interface

You can quickly manipulate and analyze business data by pointing to a menu command and pressing the mouse button. Since the Graphical Environment uses familiar menus and dialog boxes to display commands and their options, extensive training is not necessary in order to use the ad-hoc reporting and data analysis features of the software.

System Administration

The Administration display screen consolidates many system administration tasks into a single point-and-click menu system. As an added feature, many tasks contain drill-down capabilities to help you navigate to related administration tasks and information.

Message Center

This feature enables you to send, receive, and store messages. If you have a message queued to your user or station number, it is displayed when you enter the Message Center.

User Preference Options

You can select a number of options, including automatically starting PCME when you log in and displaying detailed screen tips.

Online Help

Help is available by pressing F1 or clicking on the Help button in any dialog box.

Getting Started

These topics present the basic skills you need to begin working with the Business Information Server. Topics covered in this section include

- Executing Business Information Server Software
- Using the Transmit Key
- Signing on to the Business Information Server
- Displaying the Active Screen
- Using Applications
- Function Keys

Executing Business Information Server Software

When Business Information Server software is executing, the sign-on screen is displayed. This screen indicates the level of software you are using, your station (terminal) identification number, and the Business Information Server site letter you are using. From here, you must enter a registered user-id, department number, and password (see your supervisor or Business Information Server system administrator for these), or press the **SignOn** key to gain access to the system with a user-id of NEWUSER.

```
Business Information Server
Your Company Name Goes Here
Station: n Site: x Level: y

Please enter the following information,
or press SignOn to sign on as a new user.

User-id _________
 Dept # _________
Password _________


Press XMIT when complete.

SignOn KeyHlp ReadMe Help Exit
```

When you are done using Business Information Server software, you must sign off and exit the software. Type **x** and press Transmit. See the *Command Reference.*

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If you do not see a sign-on screen, see your supervisor or system administrator.

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Linux/UNIX OS Þ Þ

Executing Software on a Business Information Server for Linux or UNIX Operating System

You can execute the software that is installed on a Business Information Server for Linux or UNIX Operating System using either of these methods:

- Use the Business Information Server log-on.
- Log on to the Linux/UNIX operating system and execute Business Information Server software as a program.

Since log-ons and options are case-sensitive on the server for the Linux/UNIX Operating System, you must enter these commands in lowercase to execute the server software.

The easiest way to execute this software is with the log-on. You enter this from the log-on prompt to execute software for the default Business Information Server site. (The default site on any host computer is the site that was installed first.)

Simply type **mapper** and press **Return**. The system displays a password prompt. Type the password you received from the System Administrator and press **Return**. After a few seconds, the system sign-on screen is displayed and you can sign on. You do not need a Linux or UNIX operating system log-on to execute the software with this method; however, you can only access the default site. To access another site, you need a Linux or UNIX operating system log-on so you can execute the system software as a program.

When executing Business Information Server software as a program, you can specify one or more of the following options:

- A specific Business Information Server site
- A user-id with which to sign on and, optionally, a run name to execute
- A specific station identification or any available station to access

First log on to the Linux/UNIX operating system with your Linux or UNIX operating system log-on. When the shell prompt is displayed, use the following format to type the command. Press Return to execute the system software.

```
mapper [-ssl -luser-
id,dept,psw,run,data1,data2,... -t³-Tdevice-name]
```

Ü Ü Linux/UNIX OS

Using the Transmit Key

Before doing anything on the Business Information Server, you must know what key to press to transmit your requests to the system. Documentation refers to this key as the Transmit key. To find which key designates the Transmit key, look at the sign-on screen. Just below the User-id, Dept #, and Password input fields, the following sentence is displayed: Press *key* when complete.

Where *key* is the Transmit key.

Windows Þ Þ

You can either press **Transmit** or double-click with the mouse pointer positioned after the information you want to transmit to the system.

Ü Ü Windows

Signing on to the Business Information Server

Before you can access the Business Information Server system, ensure you have the following information:

- User-id
- Department number
- Password (depending on the system configuration)

If you do not know this sign-on information, ask your supervisor or system administrator.

There are several ways to access your Business Information Server system, depending on the equipment and the system configuration you are using.

If you are using...	**Then...**
Workstation client	Sign-on information is entered into a workstation client script. Sign on to the Business Information Server by opening the script through the workstation client. If you leave some of the information blank, the sign-on screen appears where you must enter your user-id, department number, and

	password when the LOGON run is executed.
Character-based screen	Use the sign-on screen menu or the Sign On control line format to enter sign-on information.

Windows Linux/UNIX OS Þ Þ

To access your Business Information Server system, use the sign-on screen menu or the Sign On control line format to enter your information.

Sign-On Screen Format

```
mapper [-ssl -luser-id,dept,password[/new-
password],run,
data1,data2,... -t³-Tdevice-name]
```

The following table explains the format fields:

Field	Description
-ssl	Site option, -s, and the letter of the Business Information Server site to access, *sl*.
-l	Indication that a sign-on follows. (This is the letter L; however, it must be lowercase.) ***Note:*** The maximum number of characters allowed for the –l field is 996. However, this number includes the user-id, Dept, Password, Run, and all characters included in the data field.
user-id	User-id with which to sign on to the system.

dept	Department number.
password/new-password	Optional password/new password. If the system does not require a password and you enter any subsequent statement specifications, type two commas designating the password field.
run	Run name to execute after sign-on.
data1,data2,...	Literal data to pass to the run
-t	Indication that you are to use the first available station number that is not configured without checking the configuration report. (This t must be lowercase.)
-Tdevice-name	Indication that you are to use the station number configured as device-name. (This T must be uppercase.)

Examples

In this example, user NEWUSER signs on to department 7 of site C and the HELP command executes: mapper -sc -lnewuser,7,,help This command configures the station number tty08: mapper -sc -Ttty08

Ü Ü Windows Linux/UNIX OS

Sign-on Control Line Format

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]user-id,dept[,password/new-password,run,data1,data2,...]

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][domain\]user-id,dept[,password/new-password,run,data1,data2,...]

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The following table explains the quick reference format fields:

Field	Description
]	A right bracket, used with the control line format only. This character must begin the sign-on.
domain	An optional domain name used to qualify the user-ID.
user-id	A string of characters, usually your initials or name. The

	system administrator must register your user-id.
dept	Department number, which is a number designating a group of users. The system administrator must also register your department number.
password/new-password	A string of characters used to prevent other users from using your sign-on.
run	Run name to execute after sign-on.
data1,data2,...	Literal data to pass to the run.

Remember to sign off the system whenever you are away from your terminal, to prevent anyone else from using your sign-on.

Password Guidelines

For security purposes, it is recommended that you change your password regularly.

You can create or change your password using one of the following

- The graphical screen

  To use the graphical screen, execute the LOGON script. This script takes you through the steps for changing your password.

- The character-based screen

  To use the character-based screen, enter your sign-on information in the Sign On Control Line Format on the top line of the display along with the values in the following table in the Password field.

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Note: If OS 2200 user-ids are configured in your system, the only way to change an OS 2200 password is at the Tip Session Control or Demand "Enter your user-id/password" prompt.

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Functionality	Control Line Format
Create a new password	`]User-id,Dept#,/newpassword`

Change your password	`]User-id,Dept#,oldpassword/newpassword`
Clear your password	`]User-id,Dept#,oldpassword//`

For more information, ask your administrator about the LOGON script and any special password procedures you may have at your site.

Displaying the Active Screen

Do the following to display the active screen in the demonstration database:

Execute the Business Information Server software. With idle log-on display, press **SignOn**. You are now signed on to the Business Information Server as NEWUSER in the demonstration database.

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Press **SignOn** if it is displayed; otherwise, see your supervisor or system administrator for instructions on how to access the demonstration database at your site.

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The active screen in the demonstration database looks something like this:

```
Business Information Server
     Your Company Name
Station: n Site: n Level: n
      User-id: NEWUSER
         Cabinet: 0
```

You are now ready to begin working with the Business Information Server.

Using Applications

If you purchased an application package to use on your system, or if someone else at your site has developed an application for you, you are now ready to use it. Simply read the documentation that accompanies the application. Documentation may be online or printed; it may include installation instructions, if necessary, and it should include instructions on how to sign on to the system and use the application.

If you intend to use only the applications provided for you, you do not need to read the rest of this part of the guide; however, reading more about the Business Information Server might give you some good ideas for other uses of the system.

Function Keys

As you work in the Graphical Interface for Business Information Server or at a terminal, a set of function keys appear on the function key bar at the bottom of the Graphical Interface window or the terminal screen. The keys available on the function key bar change to fit your environment and the application you are using. You can use these function keys to perform a variety of operations. The following list describes some of the keys that you may see displayed on the function key bar.

- AddLin

  Adds a line to the report where the cursor is positioned.

- Apps

  Lists all the scripts to which you have access. You can execute one of the scripts by pressing Apps, moving the cursor to one of the script names, and transmitting.

- Edit

  Brings you into edit mode. From there you have access to the Line Change menu, from which you can choose to add, delete, or duplicate lines.

- LineCh

  Displays a menu from which you can change a certain line. For example, to duplicate a line, place the cursor on the line to be duplicated and press LineCh. Then move the cursor to the Duplicate Line selection and

transmit. This function key on the function key bar only after you press Edit.

- Paint

  Repaints the screen. For example, with a report on display, type some text but do not transmit. If you press Paint, the report is redisplayed without the text you just typed.

- Remote

  Lists systems to which you have access. You can transfer your station to another Business Information Server site by pressing Remote, moving the cursor to a system on the menu, and transmitting.

- Report

  Lists all the reports to which you have access. You can choose a report in which to work by pressing Report, moving the cursor to a report on the menu and transmitting.

- Resume

  Continues a function from the point at which it pauses or stops. For example, since the Find (F) manual function pauses after it finds the first occurrence of a character string, you must press Resume to find successive occurrences.

- Return

  Displays a previous screen. For example, if you display a report while the active screen is on display, pressing

Return then redisplays the active screen. (The screen you are returned to varies depending on the current screen display and the path you used to obtain it.)

- SgnOff

  Brings you back to the sign-on screen. From there, you can sign on to the system again or exit the system.

- SignOn

  Signs you on to the system as a new user. This gives you access to the demonstration database and the basic manual functions. Use this for training purposes only.

- SOE

  Places a start-of-entry (SOE) character at the cursor position for data updates.

  If the SOE key is not displayed on the function keys bar, use Ctrl-0 to enter an SOE character.

  SOE is not available on T27 terminals. Use DEL.

- Tasks

  Displays a selection of jobs you might want to accomplish, such as printing, comparing reports, or finding data. For example report, press Tasks, then select Print from that menu. Another menu then displays specific print commands from which you can choose.

- Undo

  Reverses the last update command performed. This key does not undo the action of a script, but it does reverse

the action of some commands (for example deleting a report). Note that you can only undo the last update operation if Undo is displayed on the function key bar.

- View

  Displays a menu of selections that enables you to control the report display. For example, to shift the report 35 characters to the left, press View, then enter 35 in the Shift menu selection and transmit.

Screens, Reports, and Results

This section covers the following topics:

- Screens
- Reports
- Results

Screens

When you display a report, a screen similar to the example is displayed.

Control Line

The first line on the screen is the control line. The control line has two fields that enable you to display different parts of the report and one field (on the far right) that denotes the report being processed. You also enter command calls on the control line. It is important to note that the control line is not a part of a report. For example, when you print or count lines in a report, the control line is not included.

Here is a typical control line:

```
 Line> Roll> 2B0
```

The Line field indicates the number of the line currently at the top of the screen. The preceding control line indicates that line 24 of the report is at the top of the screen. To bring a different line to the top, type its number in the Line field.

For example, to display line 15 at the top of the screen, type **15** in the Line field. Use the Roll field to scroll the display forward or backward. To scroll forward, type **+** or leave the field empty and press **Transmit**. To scroll backward, Type **-** and press **Transmit**. To scroll forward a certain number of lines, type **+** and the number of lines, then press **Transmit**. To scroll backward a certain number of lines, type **-** and the number of lines. For example, **-13** scrolls the display backward 13 lines.

You can shift the displayed report in other ways, too. For example, using the Display Alternate Format (F*n*) command, you can display a different selection of fields. Use the following commands to manipulate the report on the screen:

- Create Temporary Format (VIEW)

- Display Alternate Format (F*n*)
- Display and Hold Headings (DH)
- Hold Characters on Screen (HC*n*)
- Hold Lines on Screen (H*n*)
- Shift Report (S*n*)

For more information, see the *Command Reference*.

Function Key Bars

The bottom line of your screen is the function key bar. It contains the names of keys to press to perform commands and navigate through the system. Like the control line, the function key bar is not considered part of a report.

```
Paint SOE Return Tasks View Help Edit
```

Note that the function keys included in the function key bar vary depending on what you are doing. Also note that applications at your site may change the function key definitions.

The function key bar, optionally, is displayed at the bottom of the screen. If it is not, click on FKeys in the menu bar at the top of the screen to display the list of available function keys.

Function Keys

The following table describes some of the keys that you may see displayed on the function key bar. The keys available on the function key bar change to fit your environment and the application you are using.

Function Key	Description
AddLin	Adds a line to the report where the cursor is positioned.

Edit	Brings you into edit mode. From there you have access to the Line Change menu, from which you can choose to add, delete, or duplicate lines.
LineCh	Displays a menu from which you can change a certain line. For example, to duplicate a line, place the cursor on the line to be duplicated and press **LineCh**. Then move the cursor to the Duplicate Line selection and press **Transmit**. This function key appears on the function key bar only after you press **Edit**.
Paint	Repaints the screen. For example, with a report on display, type some text but do not transmit. If you press **Paint**, the report is redisplayed without the text you just typed.
Remote	Lists systems to which you have access. You can transfer your station to another Business Information Server site by pressing **Remote**, moving the cursor to a system on the menu, and transmitting.
Report	Lists all the reports to which you have access. You can choose a report in which to work by pressing **Report**, moving the cursor to a report on the menu, and transmitting.
Resume	Continues a function from the point at which it pauses or stops. For

example, since the Find (F) manual function pauses after it finds the first occurrence of a character string, you must press **Resume** to find successive occurrences.

Return	Displays a previous screen. For example, if you display a report while the active screen is on display, pressing **Return** then redisplays the active screen. (The screen you are returned to varies depending on the current screen display and the path you used to obtain it.)
Scripts	Lists all the scripts to which you have access. You can execute one of the scripts by pressing **Scripts**, moving the cursor to one of the script names, and transmitting.
SgnOff	Brings you back to the sign-on screen. From there, you can sign on to the system again or exit.
SignOn	Signs you on to the system as a new user. This gives you access to the demonstration database and the basic manual functions. Use this for training purposes only.
SOE	Places a start-of-entry (SOE) character at the cursor position for data updates. If the SOE key is not displayed on the function key bar, use **Ctrl-O** to enter an SOE character.

Tasks	Displays a selection of jobs you might want to accomplish, such as printing, comparing reports, or finding data. For an example report, press **Tasks**, then select Print from that menu. Another menu then displays specific print commands from which you can choose.
Undo	Reverses the last update command performed. This key does not undo the action of a run, but it does reverse the action of some commands (for example, deleting a report). Undo is usually displayed on the function key bar; if it is not displayed, you cannot undo the last update operation.
View	Displays a menu of selections that enable you to control the report display. For example, to shift the report 35 characters to the left, press **View**, then enter **35** in the Shift menu selection and press **Transmit**.

Example Screen

```
Line> 1 Roll> 2B0
.DATE 02 NOV 04 12:38:12 RID 2B 20 JAN 04 MAPPER
.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
IP 970924 LS BLACKBOX1 436767 84389 AMCO 970923
970924
IP 970925 LS BLACKBOX1 436768 84390 AMCO 970923
970925
IP 970919 LS BLACKBOX2 637071 84353 INTR 970918
970919
OR 970110 LS BLACKBOX4 94754 ARCO
SC 970110 LS BLACKBOX5 675281 97441 FEDS 970131
IP 970922 LS BLACKBOX5 737582 84040 AMCO 970922
970922
SH 970903 LS BLACKBOX5 746327 54237 FEDS 970901
970902
SH 970902 LS BLACKBOX6 368061 54438 FEDS 970901
970901
SH 970909 LS BLACKBOX6 777324 54232 DICO 970907
970908
SH 970903 LS BLACKBOX6 785367 52833 ARCO 970901
970902
IP 970916 LS BLACKBOX6 926581 89381 INTR 970915
970916
OR 970910 LS BLACKBOX7 99842 FEDS
OR 970927 LS BLACKBOX7 99725 INTR
SC 970108 LS BLACKBOX7 665481 97541 FEDS 970122
```

IP 970927 LS BLACKBOX7 733597 84351 AMCO 970927
970927
SH 970902 LS BLACKBOX7 744627 44232 INTR 970901
970901
IP 970915 LS BLACKBOX7 933581 84381 FEDS 970915
970915
Paint SOE Return Tasks View Help Edit

Reports

> ⚠ **Caution** Before making changes to a standard report in the demonstration database, make a copy of the report, using the Duplicate Report (XR) command, and practice on the copy. This way, you will not alter or delete part of this database for other users.

Reports are sets of related data that reside within drawers of a cabinet. Reports are organized in a structure designed for ease of use.

Example

```
.DATE 15 SEP 04 15:30:30 <-- Date line
.Sample Report <-- Report title line
*St.Status.By. Product . <-- Field heading lines
*Cd. Date .In. Type . (two shown)
*==.======.==.=========. <-- Heading divider line
IP 970824 LS BLACKBOX1 <-- Data lines
SC 970825 LS BLACKBOX1 (two shown)
```

Each report is a part of the total Business Information Server database. Commands process the data stored in the reports. The number of reports in a drawer varies. Usually, each report has fewer than 500 lines. The maximum number of lines per report may vary from drawer to drawer. See the *Command Reference* for more information about the maximum size of reports.

The Business Information Server system administrator controls the size of reports and drawers and assigns new drawers for new applications. The system software works most efficiently with a database of many short reports. You can view the data in reports anytime on your display screen, and you can position the data in a number of ways, vertically and horizontally.

Choosing specific formats enables you to view a specific selection of report fields. For example, if format two is defined as the first, third, and fourth report fields, when you request a report in that format, only those fields are displayed. The basic format, or format zero, is predefined as the leftmost columns that fit on your screen.

Windows Linux/UNIX OS Þ Þ

Each drawer can hold up to 25 report formats. You can process reports displayed in any of these formats. Your system administrator can define formats one through 25, depending on your needs.

Ü Ü Windows Linux/UNIX OS

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Each drawer can hold up to six report formats. You can process reports displayed in any of these formats. Your system administrator can define formats one through six, depending on your needs.

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Report Structure

Date Line

When a report is on display, the first line of a report is the date line, which is also called line one. The date line looks something like this:

```
.DATE 29 SEP 04 14:17:44 RID 2B 01 FEB 04 MAPPER
```

Following is an explanation of each field of the date line:

Field	Description
.DATE	Date line identifier (also called line one)
29 SEP 04	Date of last update
14:17:44	Time of last update
RID 2B	Report number and drawer letter
01 FEB 04	Date of origin of report or date report was last replaced

MAPPER User-id of last user to update report

Heading Lines

The heading lines are below the date line and contain the following types of lines:

- The report title begins in column two of the line, with a period in column one. This title is the same as the title that appears in the Report Select menu.
- Field headings begin with an asterisk (*) in column one.
- The heading divider line separates the field headings from the data. It begins with an asterisk (*) in column one and then contains equal signs (=) denoting the size of each field, with periods separating each field.

Example

```
.Sample MAPPER Report <-- Report title line
*St.Status.By. Product .Serial. <-- Field heading
line
*Cd. Date .In. Type .Number. <-- Field heading
line
*==.======.==.=========.======. <-- Heading
divider line
```

See your administrator for details on setting up heading lines for a new database.

Data Lines

The lines below the heading divider line contain the actual data of the report and are called data lines. The data lines

are commonly structured into fields. Each field is made up of one or more columns (character positions) and typically separated by tab characters. Depending on your terminal or the way your terminal is currently set up, the tab characters may or may not be visible on the screen.

Each line in the report begins with a character other than your data, such as an asterisk, a period, or a tab. These characters determine the way Business Information Server software interprets each line.

Line Types

Reports can contain different types of lines, or line types, which are used to distinguish different kinds of data. For example, the line type determines whether the line is column-formatted or whether the system edits it for allowable characters. The character in column 1 defines the line type.

If you are designing a report, use line types to specify how the system will interpret your data. Use the following descriptions to decide which line types to use. If you are using a report that someone else created, read the following descriptions of each line type to understand how the report was set up and how to use each type of line.

The leftmost character on a line determines the line type. Reports can have these kinds of line types:

Line Type	Description
Tab	Column-formatted edited lines, starting with a tab character.
Asterisk	Column-formatted nonedited lines, starting with an asterisk (*).
Period	Freeform comment lines, starting with a period (.). (If you look back at

	line one, you can see that it is a period line.)
Special	Column-formatted edited lines, starting with any valid character except a tab, asterisk, or period. R line types should not be used because they conflict with range (R) parameters on many functions.

Note: When determining which line types to use, remember that, if you use the Search (S, SRH) command, asterisk and period lines are trailer lines. This means that they are associated with the previous data line and remain with that data line when it is processed. For example, a Search result shows all tab lines found, along with any asterisk or period lines that follow each tab line. In the Search command, asterisk lines are trailer lines to tab lines, and period lines are trailer lines to asterisk lines. In other commands, the system considers all line types other than the line type being processed as trailer lines.

Tab Lines

Type a tab character in column one to specify a column-formatted, edited line. The tab character is commonly represented on the screen by a vertical bar (|) but may be displayed as a space or a small dot. Most data lines in reports are tab lines.

You can display and process tab lines in different formats. Generally, most data lines in a report are tab lines.

Asterisk Lines

Type an asterisk (*) in column one to specify a column-formatted, nonedited line. Use asterisk lines for report headings. You can also use asterisk lines for data lines. The Iterative Calculate (ICAL) command, for example, uses asterisk lines to store equations.

Asterisk lines range from 40 to 998 characters in length and can be displayed and processed in different formats.

In the Search command, asterisk lines are considered trailer lines if they follow tab lines or other asterisk lines.

Period Lines

Type a period (.) in column one to specify a freeform, nonedited line. The date line and title line of reports are period lines. You can also use period lines to include comments in the report, since Business Information Server software does not interpret these lines as data. Period lines do not shift from left to right when you manipulate the report display.

Period lines are restricted to 80 characters in length. Since period lines do not shift, if your terminal is only 80 characters wide, you will only see 80 characters of the data.

In the Search command, period lines are considered trailer lines if they follow asterisk or tab lines.

Special Lines

Special lines are typically only used in advanced applications. To specify a special line, type any character other than a tab, period, or asterisk in column one to specify

a column-formatted, edited line. Special lines have all the other characteristics of tab lines.

Demonstration Database

The demonstration database consists of reports on which you can practice commands. See your supervisor or system administrator for instructions on accessing the demonstration database at your site.

Results

When you perform certain commands, a result is displayed showing the outcome of the operation. The word RESULT appears at the far right of the control line.

```
Line> 1 Roll> - RESULT
.DATE 02 NOV 04 12:38:12 RID 2B 20 JAN 04 MAPPER
.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
IP 970924 LS BLACKBOX1 436767 84389 AMCO 970923
970924
IP 970925 LS BLACKBOX1 436768 84390 AMCO 970923
970925
IP 970922 LS BLACKBOX5 737582 84040 AMCO 970922
970922
IP 970927 LS BLACKBOX7 733597 84351 AMCO 970927
970927
OR 970917 LS BLACKBOX9 98755 AMCO
SH 970904 LS BLACKBOX9 714577 64231 AMCO 970901
970903
SH 970906 LS GREENBOX7 669624 54682 AMCO 970901
970905
SC 970110 LS GREENBOX8 975481 99943 AMCO 970110
OR 970310 LS GREENBOX9 99951 AMCO
..... END REPORT .....
```

Result

A result is a temporary report produced by some system functions after they finish processing. When initially created, a result always has a report number of -0. The maximum lines per result may vary from drawer to drawer. See the *Command Reference* for more information about the maximum size of results.

This result is lost as soon as you display another report, or sign off the system. If you want to keep the result, you can do one of the following:

- Place the result into a permanent report, using the Replace Report (REP) command.
- Duplicate the result into a permanent report, using the Duplicate Report (XR) command.

You can save the result temporarily, using the Save Report Version (SV) command.

In addition to printing, duplicating, and replacing results, you can perform many of the other commands available in the Business Information Server. To execute a manual function against a result on display, use a minus sign (-) in place of a report number.

Update Results

An update result is also a temporary copy of a report. The difference between a result and an update result is that you can use an update result to modify the original report. Like results, update results can be updated without disturbing the original report. An update result, however, is most useful when it contains data or modifications that you do want to insert into the original report.

Use the following commands to modify the original report, using lines from the update result:

- Update (UPD) copies the lines in the update result into their original positions within the original report. Any modifications to the lines in the update result are carried to the new report.
- Delete (DEL) deletes the lines in the update result from the original report.
- Extract (EXT) deletes the lines in the update result from the original report and converts the update result to a result, which you can save as a permanent report.

The following commands create update results:

- Calculate Update (CALU)
- Change (CHG) with the OU option
- Locate (LOC) with the OU option
- Match Update (MAU)

- Search Update (SU)

Managing Reports

This section covers the following topics:

- Creating a Report
- Opening a Report
- Copying a Report
- Saving a Report
- Closing a Report
- Deleting a Report
- Viewing a Report
- Moving Through a Report
- Updating a Report
- Printing a Report

Creating a Report

To create a new report

1. In home position, or on the control line (if you already have a report on display), type **ar**.
2. Press **Transmit**.
3. In the Report or Drawer field, type the drawer letter in which to create the new report.
4. Tab to the **Title field** and type a title for the new report.
5. Press **Transmit**.

The system creates a new, blank report. Write down the new report identifier or plan to access this report by title using the RL command or the Report key.

See the *Command Reference* for more information.

Opening a Report

There are a number of ways to open a report. One way is to type a report identifier that includes a report number, a drawer letter, and a cabinet number. Another way is to press **Report** to display a list of available drawers; highlight the desired drawer (tab to it or click on it) and press **Transmit**; highlight the desired report and press **Transmit**.

Note: If you do not include a cabinet number as part of the report identifier, the system uses the current cabinet as a default.

The demonstration database contains the following reports:

- 1B or 1B0
- 2B or 2B0
- 1C or 1C0
- 1D or 1D0

To keep data secure on the Business Information Server, there are a number of security features that protect reports from unauthorized display and updating. For more information, see Accessing Drawers or Protecting Reports.

To display report 2B

1. Position the cursor in home position (upper left corner of the screen). Most keyboards have a Home key you can use for this purpose.
2. Type **2b**.

3. Press **Transmit**.

Report 2B is displayed.

For more information on specifying reports to display or process, see the *Command Reference.*

Copying a Report

To copy an existing report

1. Display a report you wish to copy, such as 2B.
2. On the control line, type **xr.**
3. Press **Transmit**.
4. In the Title field, type a title for the report, such as Practice Report.
5. Press **Transmit**.

The system copies the report. Write down the new report identifier, which you can find in the middle of the second line on the screen. You can use this identifier to access the report later. You can also use the RL command or the Drawer key (found on the function key bar) to access the report by looking for the title you gave it when the list of reports is displayed.

You may use this copied report when trying out the commands described in the rest of this guide.

See the *Command Reference* for more information.

Saving a Report

To save a copy of a report (or result), follow the instructions under Copying a Report.

To replace a report with a result (or a different report)

Note: You are not allowed to replace a report unless you were the last person to update that report. A system message appears after you press **Transmit** if you were not the last person to update the report.

1. With the result or report to save on display, type **rep**.
2. Press **Transmit**.
3. In the Report field, type the report identifier of the report to be replaced. To replace a nonexistent report, type a new report identifier.
4. Type a title for the report (optional).
5. Press **Transmit**.

The system replaces the report you specified in the input screen with the result or report on display.

To undo the replace, press **Undo** immediately after performing the replace, before doing anything else on the system.

Note: If Undo is displayed on the function key bar, it is possible to undo the previous transaction. You can use Undo in many updating situations, not just after replacing a

report. See the *Command Reference* for complete information on these commands.

Closing a Report

To close a report (release the screen)

1. On the control line, type ^ (a caret).
2. Press **Transmit**.

The active screen is displayed.

Other ways to release the screen are to

- Press **Quit** if it is displayed on the function key bar.
- Press **Return** until the active screen is displayed.

For More Information

You have just used the control line and a few commands to manipulate the screen display. For more details on these and other methods of manipulating the screen, see the following commands in the *Command Reference*:

- Create Temporary Format (VIEW)
- Display Alternate Format (F*n*)
- Display and Hold Headings (DH)
- Hold Characters on Screen (HC*n*)
- Hold Lines on Screen (H*n*)
- Line Control (L)
- Release Display (^)
- Shift Display (S*n*)

Deleting a Report

To delete a report

1. Display the report you wish to delete.
2. On the control line, type **dr**.
3. Press **Transmit**.
4. In the Report field, type the report number and drawer letter of the report to delete (the same report that was on display).
5. Press **Transmit**.

The system deletes the report.

Note: You cannot delete a report unless you are the owner of the report.

See the *Command Reference* for complete information on these commands.

Viewing a Report

When you display a report, a screen similar to the following is displayed: Line> 1 Roll> 2B0

```
.DATE 02 NOV 04 12:38:12 RID 2B 20 JAN 04 MAPPER
.Production Status Report Corporate Production B000002
*St. Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code. Plan .Actual.
*==.=======.==.=========.======.======.=
====.====.======.======.
IP 970924 LS BLACKBOX1 436767 84389 AMCO 970923
970924
IP 970925 LS BLACKBOX1 436768 84390 AMCO 970923
970925
IP 970919 LS BLACKBOX2 637071 84353 INTR 970918
970919
OR 970110 LS BLACKBOX4 94754 ARCO
SC 970110 LS BLACKBOX5 675281 97441 FEDS 970131
IP 970922 LS BLACKBOX5 737582 84040 AMCO 970922

970922
SH 970903 LS BLACKBOX5 746327 54237 FEDS 970901
970902
SH 970902 LS BLACKBOX6 368061 54438 FEDS 970901

970901
SH 970909 LS BLACKBOX6 777324 54232 DICO 970907
970908
SH 970903 LS BLACKBOX6 785367 52833 ARCO 970901
970902
IP 970916 LS BLACKBOX6 926581 89381 INTR 970915
970916
OR 970910 LS BLACKBOX7 99842 FEDS
OR 970927 LS BLACKBOX7 99725 INTR
```

SC 970108 LS BLACKBOX7 665481 97541 FEDS 970122
IP 970927 LS BLACKBOX7 733597 84351 AMCO 970927

970927
SH 970902 LS BLACKBOX7 744627 44232 INTR 970901

970901
IP 970915 LS BLACKBOX7 933581 84381 FEDS 970915

970915

1 2Paint 3SOE 4Return 5 6Tasks 7View 8Help 9 10Edit The top line of this screen is called the control line. This is where you position the cursor to type requests you wish to send to the Business Information Server. You can type requests, such as 2b to display report 2B, after Line? or after Roll?. Make sure the cursor is resting either on a space or on the last character of your request when you press Transmit, because the character on which the cursor is resting is sent to the system for processing, and if you are not careful, you could send a character you did not intend to send.

For more information on Business Information Server reports and screens, see Screens, Reports, and Results.

Moving Through a Report

This procedure is most useful for reports wider than the screen.

Changing Your View of the Report Horizontally

Note: You must use a workstation to view all of the fields of a wide report.

To change the horizontal view

1. On the control line, type **view**.

2. Press **Transmit**. A screen is displayed that looks like the report, only with blank fields. This is called a function mask.

3. Tab to the first field you wish to view.

4. When you tab, the cursor moves to different positions beneath the line that contains mostly equal signs (=). This blank line is where you enter the instructions, called parameters, to indicate the fields you want to process and how they are to be processed. See Filling Out Function Masks for more information on filling out function masks.

5. In the line below the equal signs, type **x** in the first character position of the field you wish to view.

6. Repeat steps 3 and 4 for all the fields you wish to view.

7. Press **Transmit**.

The fields you selected are displayed.

To return to the original view

1. On the control line, type **f0**.

2. Press **Transmit**.

The leftmost columns of the report are displayed.

Changing Your View of the Report Vertically

To scroll down

1. Position the cursor after Roll? on the control line and type the number of lines you wish to scroll down.

2. Press **Transmit**.

To scroll up

1. Position the cursor after Roll? on the control line and type a minus sign (**-**) followed by the number of lines you wish to scroll up.

2. Press **Transmit**.

To display the report at a certain line number

1. Position the cursor after Line? and type the line number of the report at which to start the display.

2. Press **Transmit**.

To return to the top of the report

1. Position the cursor after Line? and type a minus sign (**-**).

2. Press **Transmit**.

Updating a Report

Adding Blank Lines

To add blank lines in a report

1. Position the cursor on the line above where you want to add a line.

2. Press **Edit** if it is displayed in the function key bar; otherwise, go to step 3.

3. Press **Addlin**.

If neither Edit nor Addlin is displayed, type **looksw** in the control line and press **Transmit** to display the menu interface to the system.

The system adds one blank line.

Note: Remember to press Undo if you are practicing on the demonstration database.

To add more than one blank line

1. Position the cursor on the line above where you want to add lines.

2. Press **Edit** if it is displayed in the function key bar; otherwise, go to step 3.

3. Press **LineCh**.

4. Type the number of lines to add in the Add lines field.

5. Press **Transmit**.

The system adds the number of blank lines you requested. Remember to press Undo.

To add a blank line using a Graphical Presentation Client

1. Position the cursor after the line you want the blank line to follow.

2. From the Edit menu select Add Line(s).

See the *Command Reference* for more information.

Deleting Lines

To delete one line

1. Position the cursor on the line to delete.
2. Press **Edit** if it is displayed in the function key bar; otherwise, go to step three.
3. Press **Delete**.

The system deletes one line. Remember to press Undo.

To delete more than one line

1. Position the cursor on the first line to delete.
2. Press **Edit** if it is displayed in the function key bar; otherwise, go to step three.
3. Press **LineCh**.
4. Type the number of lines to delete in the Delete lines field.
5. Press **Transmit**.

The system deletes the number of lines you requested. Remember to press Undo.

To delete lines using a Graphical Presentation Client

1. Position the cursor before first line to delete.
2. Highlight the all of the lines to delete.

3. From the Edit menu select Delete Line(s).

See the *Command Reference* for more information.

Copying Lines

To copy one line several times at the present location

1. Position the cursor on the line to copy.
2. Press **Edit** if it is displayed in the function key bar; otherwise, go to step three.
3. Press **LineCh**.
4. Type the number of times to copy the line in the Duplicate line field.
5. Press **Transmit** .

The system copies the line the number of times you requested. Remember to press Undo.

To copy lines using a Graphical Presentation Client

1. Position the cursor before the line to copy.
2. Highlight the line(s) to copy.
3. From the Edit menu select Copy.
4. Position the cursor at the location to place the lines.
5. From the Edit menu select **Paste**.

See the following topics in the *Command Reference* for more information

- Move Line

- Put Line
- Yank Line

Printing a Report

The printer you wish to use must be properly configured in the Business Information Server.

To print on a system printer

1. Display the report or result to print.
2. On the control line, enter **pr**.
3. Press **Transmit**.
4. Fill out the Print input screen. If you need help filling out the fields, press **Help** with the cursor positioned in the field you would like more information on.
5. Press **Transmit**.

The system prints the report or result on the system printer. This can mean many different things depending on the kind of Business Information Server you are using and how your site is set up. Find out the location of the printer from your supervisor or system administrator.

To print on an auxiliary printer

1. Display the report or result to print.
2. On the control line, enter **cop**.
3. Press **Transmit**.
4. Fill out the input screen.

5. Press **Transmit**.

6. If the printer you requested is connected to your workstation, press **SgnOff**.

Note: The system prints the report or result on the printer you requested. See your supervisor or system administrator if you have problems printing.

See the *Command Reference* for complete information on these commands. See also the Auxiliary (AUX) command for more information on controlling printing.

Managing Data

This section covers the following topics:

- Locating Data
- Changing Data
- Sorting Data
- Calculating Data

Locating Data

To locate data in a report

1. Display a report in which you want to locate data.
2. On the control line, type **loc *string,*** where *string* is the data you want to locate. For example, type **loc amco.**
3. Press **Transmit**.

The report is displayed at the first occurrence of *string.* To find more strings, press **Resume** (normally F1).

Note: To locate a report identifier, enclose the string in quotation marks. For example, '2B0'.

You can also use the Search (S), Find (F), or Binary Find (BF) commands to locate data. Use these commands when you want to find data in a certain column of the report or across more than one report. The S, F, and BF commands each have unique features to meet your needs in different situations and with different database structures.

The *Command Reference* provides complete information on these commands. If you have special needs for locating data, you should read the Description and Outcome sections of the commands, scan the Options and Parameters sections, and look through the examples to see if the situations described match your situation.

Changing Data

To change data in a report

1. Display a report in which you want to change data.

2. On the control line, type **chg ;/*target*/*replacement*/**, where *target* is the data you want to change, and *replacement* is the new, changed data. For example, type

   ```
   chg ;/blackbox/mauvebox/
   ```

   Before pressing **Transmit**, be sure the cursor does not rest on an unwanted character that will be sent to the system for processing.

3. Press **Transmit**.

A result is displayed that includes all lines, with the target strings changed to the replacement string. This result is temporary. If you display another report, release the screen, or sign off the system, it no longer exists. See Results for more information.

Some of the other ways to change data include

- Manually updating the report with changes.

- Using arithmetic types of commands that perform calculations and place results into fields of the report, creating a result that you can save if you wish.

- Using updating functions or options to create update results. You can modify update results and then blend

the changes back into the original report. For more information on updating functions and update results, see Results.

- Using the Totalize (TOT) command to fill fields with data. This operation is especially handy to use in conjunction with one of the updating commands, such as Search Update (SU). You can search for data you wish to change to something else, then use the TOT command to fill the fields with the changed data. Finally, you use the Update (UPD) command to blend those changes back into the original report.

See the *Command Reference* for complete information on this command.

Sorting Data

To sort data in a report

1. Display the report you wish to sort.
2. On the control line, type **sort**. A function mask is displayed.
3. Tab to the field you wish to sort in alphabetical order.
4. In the line below the equal signs, type **1** in the first character position of the field you wish to sort.
5. Move the cursor to the next blank line.
6. Press **Transmit**.

A result is displayed with the field you selected sorted in alphabetical order.

See the *Command Reference* for complete information on the SORT command.

Calculating Data

To add numbers vertically in a field

1. Display the report or result containing the data to process.
2. Type **tot**.
3. In the function mask, type **+** in the field to add.
4. Move the cursor to the next blank line.
5. Press **Transmit**.

A result is displayed. The sum of the values in the selected field appears at the bottom of the result.

See the *Developer's Guide* for a brief description of all the different commands available for performing calculations.

Then, in the *Command Reference*, look at commands such as Calculate (CAL), Arithmetic (A), and Count (CNT). Read the first few screens or pages of each command to see what capabilities each one offers. Also, check the examples to see if the situations described match your situation.

See the *Command Reference* for complete information on the Totalize (TOT) command.

Sending and Receiving Messages

This section covers the following topics:

- Sending Messages
- Receiving Messages

Sending Messages

To send an entire report to another workstation

1. Display the report or result to send.
2. On the control line, type **send** *station-number,* where *station-number* is the number of the workstation to which to send the report.
3. Press **Transmit**.

The system sends the report to the other station.

You can also use the Send Report to User (SNU) and Station-to-Station Message (SS) commands to send messages to other users.

2200 Þ Þ

You can also use the CALL command to have an interactive call session between users.

Ü Ü 2200

Receiving Messages

If you are using Graphical Interface for Business Information Server, your station signals an incoming message with a single beep when you press **Transmit** or enter a caret (^). Note that the beep does not sound again until you press **Transmit** or enter a caret (^).

To display a message from another user

Linux/UNIX OS Þ Þ

Your terminal beeps, with MSG displayed in the upper right corner of your screen.

Ü Ü Linux/UNIX OS

2200 Þ Þ

Your terminal beeps, with MSGW displayed in the lower right corner of your screen.

Ü Ü 2200

Your terminal indicates that a message has been sent from another user.

1. Type **msg**.
2. Press **Transmit**. The message is displayed.

Note: If the message came from the local site, you can press the **Message key** to display the message. If the message came from a remote site, press **Message** to stop the beeping and enter **msg** to display the message.

Windows Linux/UNIX OS Þ Þ

To display the message following the one displayed, request the MSG command again.

Ü Ü Windows Linux/UNIX OS

2200 Þ Þ

To display the message following the one displayed, first acknowledge the current message, then request the MSG command again.

Ü Ü 2200

To acknowledge a message

1. On the control line, type **ok**.
2. Press **Transmit**.

The system removes the message from your message queue.

See the *Command Reference* for complete information on the Send Report (SEND), Message Waiting (MSG), and Acknowledge Message (OK) commands.

Communicating Requests to the Business Information Server

This section covers the following topics:

- Executing Commands
- Input Screens
- Function Masks

Executing Commands

There are two separate kinds of commands you may be executing: scripts and manual functions.

Scripts are actually a series of statements that automatically start a sequence of equivalent manual functions. Scripts also include other statements that control the logic of the script, and so on.

Since scripts are written by literally thousands of different designers around the world (you might be one of them), each script might be set up to start in a different way. Some scripts, for example, require information to be entered along with the name when you execute the script. Other scripts allow you either to go through menus or to send the information along when you invoke the run. Many designers also provide online help for their scripts. This information is specific to that application and is often accessible by typing **help** after the name or by pressing the Help function key.

This topic focuses on starting a manual function. You can request a function in one of two ways:

- Navigating through a menu path
- Entering a command, also called a function call, on the control line

Once you request a manual function, you may need to complete an input screen and function mask for the command to complete its execution. See Input Screens and Function Masks.

To request a manual function through a menu path

The following example shows how to request the Totalize (TOT) manual function:

1. Press the **Task** key.

   The Select Task menu is displayed.

2. Move the cursor to the Perform math operations selection and press **Transmit**.

   The Perform Math Operations menu is displayed.

3. Move the cursor to the Totalize data selection and press **Transmit**.

   If a report was on display, the function mask is displayed. If a report was not on display, the input screen is displayed. In some cases neither an input screen nor a function mask are displayed. See Input Screens for more details.

To request a manual function on the control line

The following example shows how to request the TOT function on the control line:

1. Place the cursor in the home position or in any control position in the control line.

2. Type **tot**.

3. Erase the rest of the line by pressing the **Erase to End of Line key** and press **Transmit**.

If a report was on display, the function mask is displayed. If a report was not on display, the input screen is displayed. In

some cases neither an input screen nor a function mask are displayed. See Input Screens for more details.

Many functions have more than one command format. Each command format serves a different purpose. For example, if you want to perform the TOT function on report 2B in cabinet zero, and that report is not on display, you can enter **tot 2b0** rather than **tot**.

Note: If a system message erases the control line, press **Home** and enter **L** to restore it. You can also press the **Paint key** if it is present on the function key bar.

Input Screens

An input screen is a screen that is displayed when you request a function. The form contains fields in which you specify how you want to use the function. Some fields are automatically filled with default values. This input screen is displayed when you request the Index User (IU) function. Note that the User-id field is automatically filled in with a default value. You can either use this value or change it by typing over it.

```
                Index User Request

     Number of lines from each report _________
                  Drawer _________
            User-id (ALL for all) NEWUSER
     Last update start date (DDMMMYY) _________
      Last update end date (DDMMMYY) _________
               First report _________
                Last report _________
```

To complete the form after you call a manual function

1. Type information in each field you want to fill in or change.

2. When you have completed the input screen, press **Transmit**.

Note: Some fields require a value. If you leave a required field blank, a system message is displayed.

Many commands allow you to skip the input screen and use a control line format. This format, which you enter on the control line, lets you specify many of the same values, as

does the input screen. For the IU command, you could enter **iu 5d** to display five lines from each report in drawer D.

Purpose of Input Screens

Input screens are displayed because the system needs more information to process your request. An input screen may be displayed if one of the following is true:

- You request a manual function while no report is on display.
- You request a manual function that needs more information than just the report name and format number.
- You have a report on display and you call a manual function that can process an entire drawer, but you enter only the function call (for example, you enter **i** rather than **i b**).

Obtaining Help

The following help is available for you when filling out an input screen:

- For an explanation of a field and its possible values, tab to that field and press **Help**. A brief description is displayed.
- For additional information about how the function works, press **Help** again. The help topic for the function is displayed.

Screens Displayed after an Input Screen Is Complete

After you complete an input screen, the next screen displayed varies depending on the command and what was on display.

- Some manual functions execute immediately when you transmit the completed input screen.

- Some manual functions require more information about the report fields to process. In these cases, a function mask is displayed. See Function Masks.

- If the system cannot execute the command based on the information you supplied, it displays a system message. For example, if you did not fill in a required field or you specified a report that does not exist, the system displays a message and does not execute the command.

Function Masks

A function mask looks like a report with blank fields. You use it to specify options, field widths, and parameters whenever you request a function that processes individual fields or character positions in a report. The field headings displayed in the mask are those that were displayed when you requested the function. If you process an entire drawer, the system displays headings from report zero.

Options are usually designated by single letters or symbols. You use them to change the way a function works. Type options in the line just above the report headings from left to right, beginning in column one.

Field widths are denoted by a line of asterisks below the heading divider line, making it easier to select the fields to process when entering function parameters. Parameters indicate which operations to perform and the order in which to perform them. They also let you select the types of lines to process.

Where to Type Options and Parameters

The descriptions and arrows to the left of this partial function mask point to the places where you type options and parameters as well as where you change the field width, if necessary.

```
Type options SEARCH
above the mask -->
.Production Status Report
*St.Status.By. Product .Serial.Produc.
*Cd. Date .In. Type .Number. Cost .
Change field
*==.======.==.=========.======.======.
width if necessary--> ** ****** ** *********
****** ******
Type parameters -->
below the mask
```

Obtaining Help

Whenever a function mask is on display, you can press **Help** to see a summary of available options and parameters for the function you are using. Press **Help** again for more details on the function.

Specifying Options

Enter options to give more detailed instructions about how to process your request. Since options vary by function, see specific functions for available options.

After you have requested a function and the function mask is displayed, type one or more options on the blank line just above the heading line in the mask. When you use more than one option, type them next to each other with no intervening spaces or commas.

Two options, D and H, are typed in this function mask:

```
  SEARCH
dh
.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
** ****** ** ********* ****** ****** ***** ****
****** ******
```

Specifying Parameters

Use parameters to indicate which fields to process and how to process them. Since parameters vary by function, see specific functions for available parameters. After you request the function and the function mask is displayed, perform the following procedure:

To type parameters on the function mask

1. Tab to the field below the asterisk line where you want to enter a parameter.

2. Type the parameter.

3. If more parameters are necessary, repeat steps one and two.

4. Move the cursor to the next blank line and press **Transmit**.

For functions that locate data, a parameter would be the character string to locate. For functions that perform math operations, a parameter would be +, -, and so on. The parameter IP has been typed in this function mask. When executed, this function mask searches for occurrences of IP in the St Cd field:

```
  SEARCH

 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
```

Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
** ****** ** ********* ****** ****** ***** ****
****** ******
ip

Processing Partial Fields

Sometimes you may need to process only part of a field. For example, you might need to search only the third and fourth columns of a six-column field. You can specify partial fields to be processed in two ways. The first procedure is useful for any function mask; the second is useful only for the functions that search reports.

To process partial fields (manual functions)

1. Types spaces over, or erase, the asterisks in the columns not to be processed. Note that you only need to erase the asterisks immediately before and after the columns that are to be processed, not every asterisk in the mask.

2. Type parameters below the appropriate asterisks.

In the following function mask, the second and fifth asterisks of the Status Date field are erased. This means that the third and fourth characters are to be processed. The parameter is typed below the asterisks of the columns to be processed.

```
  SEARCH

 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
 Plan .Actual.
 *==.======.==.=========.======.======.=====.====.
 ======.======.
```

```
** * ** * ** ********* ****** ****** ***** ****
****** ******
12
```

To search reports

1. Type a slant (/) in the parameter line preceding the column where you want the function to start processing in the field. You can also type a slant following the last column to be processed in the field.

2. Type the string to locate in the remaining portion of the field. Here, in the St Cd field, column 1 is processed; in the Status Date field, columns 3 and 4 are processed:

```
  SEARCH

.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
** ****** ** ********* ****** ****** ***** ****
****** ******
s/ /12/
```

Selecting the Type of Lines to Process

Indicate the type of lines to process in the first column of the line under the asterisk line. For example, in this function mask, since the first character below the asterisk line is a tab (represented here by a vertical bar), the function processes tab lines only.

```
  SEARCH

 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
 Plan .Actual.
 *==.======.==.=========.======.======.=====.====.
 ======.======.
 ** ****** ** ********* ****** ****** ***** ****
 ****** ******
 |
```

To change the type of line to process, type over the line-type character in column 1 with the character that represents the type of line to be processed. For example, to process asterisk lines rather than tab lines, type an asterisk in column 1 of the line below the asterisk line. Some functions have options that change the type of lines to process. For more information, see the individual manual functions in the *Command Reference.*

Using Double Function Masks

Functions that process two reports at a time display two function masks on the screen, as shown in the following example.

The top mask is from the report you requested as a second report to process, referred to by some functions as the issuing report. The lower mask is from the current report, referred to by some functions as the receiving report.

```
  MATCH
 .Factors Base Report Corporate Factors Base
 C000004
 * Product . Sub .Produc. Whole . Retail . Sales
 .Space. Demo .
 * Type . Key . Cost . Sale$ . $$$$ .Commiss. Req
 .Quantity.
 *=========.=====.======.=======.========.=======.
 =====.========.
 ********* ***** ****** ******* ******** *******
 ***** ********

 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
 Plan .Actual.
 *==.======.==.=========.======.======.=====.====.
 ======.======.
 ** ****** ** ********* ****** ****** ***** ****
 ****** ******
```

To fill in the double function mask, enter function options above the top mask and parameters below each mask, according to the function you are using. For more information, see the individual manual functions in the *Command Reference.*

Fields Displayed in the Function Mask

By default, function masks display, from left to right, the field headings of the report you requested to process. You can choose a different selection of fields in the following ways:

- Before requesting the manual function, display the report in a different format or change the user view of the report. Formats and views are variations on the way the report is displayed. To display the report in another format, press **View** and use the function mask, or use the Display Alternate Format (F*n*) command to switch to a specified format. For more information, see the *Command Reference.*

- When you initially request the manual function, indicate another format either in the control line format or in the input screen. For more information, see the individual manual functions in the *Command Reference.*

Examples

To search for multiple character strings

With some functions, you can look for multiple strings. In this function mask, data lines are found containing SH in the St Cd field and a 12 in the Status Date field:

```
  SEARCH
 dh
 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
 Plan .Actual.
 *==.======.==.=========.======.======.=====.====.
 ======.======.
 ** * ** * ** ********* ****** ****** ***** ****
 ****** ******
 sh 12
```

In this function mask, data lines containing SH or IP in the St Cd field are found:

```
  SEARCH
 dh
 .Production Status Report Corporate Production
 B000002
 *St.Status.By. Product
 .Serial.Produc.Order.Cust.Produc.Produc.
 *Cd. Date .In. Type .Number. Cost .Numbr.Code.
 Plan .Actual.
 *==.======.==.=========.======.======.=====.====.
 ======.======.
```

```
** ****** ** ********* ****** ****** ***** ****
****** ******
sh
ip
```

To search for a range of strings

Some functions let you search for a range of character strings. To search for a range of strings, call the Search (S) function (for example), display the function mask, then

1. Type the characters for the lower end of the range in the first line under the mask in the correct field.

2. Type **r** in the first column of the next line.

3. In the same line as the R parameter, type the characters for the higher end of the range in the same field as the lower end.

4. Press **Transmit**. In this function mask, 890520 is at the lower end of the range, 890803 is at the higher end. Note that the R parameter is entered in column 1.

```
  SEARCH
dh
.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
** ****** ** ********* ****** ****** ***** ****
****** ******
```

```
890520
r 890803
```

To use arithmetic operators

Some functions use arithmetic operators for parameters. In this Totalize (TOT) function mask, the plus sign vertically adds the values in the Produc Cost field:

```
  TOTALIZE

.Production Status Report Corporate Production
B000002
*St.Status.By. Product
.Serial.Produc.Order.Cust.Produc.Produc.
*Cd. Date .In. Type .Number. Cost .Numbr.Code.
Plan .Actual.
*==.======.==.=========.======.======.=====.====.
======.======.
** ****** ** ********* ****** ****** ***** ****
****** ******
+
```

To join fields in the mask

This example temporarily joins two fields to create one field that is large enough to hold the result of a horizontal calculation. Type asterisks in the spaces in the line of asterisks. The fields become one field in the result, which you can print or duplicate (do not replace the original report because of the changed fields). Note the added asterisk between the Ship Order and Spc Cod.

```
  TOTALIZE

.Production Status Report Corporate Production
B000002
```

```
*St.Serial.Produc.Order.Cust.Produc.Produc.Ship
.Spc.
*Cd.Number. Cost .Numbr.Code. Plan
.Actual.Order.Cod.
*==.======.======.=====.====.======.======.=====.
===.
** ****** ****** ***** **** ****** ******
*********
+ + =
```

Security Features

This section describes features that protect the database from unauthorized users. This section contains information about the following topics:

- Accessing Drawers
- Protecting Reports
- Script Registration

Accessing Drawers

In a typical Business Information Server, each user has access to specific drawers, depending on the permissions associated with their user-id. To achieve this, the system administrator sets up security groups that allow you access to the drawers you require, letting you read and update reports or just read reports. When you perform the following manual functions, the system checks your security group to ensure you are a valid user to access that database: Windows Linux/UNIX OS Þ Þ

- A Cabinet Switch (C or CS) command changes your current cabinet. You must have Read access to at least one drawer in a cabinet in order to switch it.

Ü Ü Windows Linux/UNIX OS

- Display Report (D) and other manual functions when using the *rdc* format for displaying or processing reports, where *rdc* is the report number, drawer letter, and cabinet number (for example, 2B0).

Protecting Reports

Reports can be protected with cabinet, drawer, and report passwords.

2200 Þ Þ

Cabinet Passwords

The system administrator assigns cabinet passwords. To access a cabinet other than the one assigned to your user-id, you can use the C or CS commands and enter the cabinet password. Your supervisor can supply you with the passwords for cabinets you need to access. For information on cabinet switching, see the *Command Reference.*

Drawer Passwords

A drawer password prevents an unauthorized user from displaying or updating any report in the drawer. You can assign or reset the drawer password for any of your drawers. You must obtain the password from your supervisor and use the Drawer Password (DPW) command to gain access to the protected drawer. For information on drawer passwords, see the *Command Reference.*

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Report Passwords

You can restrict read or write access to a report by creating a password, using either the Read Password (RPSW) or Password (PSW) command. You must then enter the password to display or update the restricted report. For information on read and write report passwords, see the *Command Reference.*

Encoded Reports

You can protect your reports by using the Encode Report (ENCODE) command, which transforms report data into unreadable code. For more information, see the *Command Reference.*

Script Registration

The administrator must register any script before you can execute it. The administrator can impose limits on the script, such as the cabinets it can access, who is allowed to start the script, or the time period in which it can execute. See your system administrator for additional information.

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In addition to registering your script to access certain cabinets, the system administrator can register your scripts to use specific drawers. The administrator can also specify certain security groups, providing security down to the drawer level by user.

Ü Ü Windows Linux/UNIX OS

Iterative Runs

This section contains the following information about iterative runs:

- Overview of Iterative Runs
- Using Iterative Runs
- Iterative Run Exercise

Overview of Iterative Runs

Iterative runs are utilities that enable you to perform a manual function, save the completed function mask, alter the mask, and use it again and again. The iterative runs are
IBFN ICNT IFND ISRH
ICAL IDAT ISOR ITOT

With the iterative runs, you can use any functionality available with the corresponding manual function, but with greater ease. For example, you can

- Save completed function masks and use them again.
- Request previously saved masks and alter them without repeatedly filling in entire masks.
- Translate function masks into their run statement formats, which can be easily added to an existing run.

Using Iterative Runs

You can use either the menu procedure or the control line format to execute iterative runs. The report you intend to process must reside in the current cabinet.

Format

`IBFN[*] report|drawer [,f,maskname,maskrep]`

The following table explains the format fields.

Field	Description
*	An asterisk to specify that field names are to be used in place of column character positions when a run statement is produced.
report	Report to process. For more details, see the *Command Reference.*
drawer	Drawer to process. Use this field to process all reports in a given drawer. For more details, see the *Command Reference.*
f	Report format in which to process data (enables you to process data in fields beyond column 80 if you do not already have those columns on display).
maskname	The name of an existing function

mask, a new function mask, or blank. (If you leave it blank, you name the new function mask when you save it.) The name can be one to 12 characters; it must begin with a letter and contain no spaces.

maskrep Report containing previously saved function masks. Use this subfield when the function masks are in a report other than the report you are processing.

Note: If you leave the *maskname* subfield blank, you cannot specify a report in the *maskrep* subfield.

Outcome

Executing the iterative runs cause the following actions to occur:

- The iterative run performs the corresponding manual function. For example, the ICAL run performs the computations just as the Calculate (CAL) function does.
- If you save the mask, the iterative run places it in the current report or another report you specify. The system inserts the completed mask as a set of asterisk lines at the end of the report. The run updates the report itself; it does not create a result.
- If a result was on display when you started the iterative run, the run places the mask at the end of the result. (To save the masks permanently, remember to replace or duplicate the result, creating a permanent report.)
- After performing the command, use the keys displayed on the function key bar, as described in the following table:

Function Key	Description
Mask	Redisplays the function mask, enabling you to change the options and parameters and perform the command again.

Report	Redisplays the original report or result.
BackUp	Displays the mask used just before the current mask. Pressing **BackUp** up to three times cycles back through the three most recently used masks. Pressing it a fourth time displays the most recent mask.
Save	Displays a menu in which you type the name of the mask to be saved and, optionally, the report to save it in. (If you use the name of a previously saved mask, the current mask replaces that mask.)
@xxx	Displays the run statement created from the most recently processed mask. For example, when using the ISOR run, you can press the **@SOR key** to obtain the SOR statement generated by the options and parameters in the mask.
Help	Displays online help for the function mask, field, or the entire iterative run you are using, depending on the screen that is on display when you press **Help**.
Quit	Terminates the iterative run and displays the active screen.

Guidelines

Follow these guidelines when using iterative runs:

- You cannot save a mask until after you perform the corresponding manual function by transmitting from the function mask.

- You can save several masks in the same report, each with a different name. You can include a maximum of six parameter lines in each mask. For example, when using ICAL, you can include as many equations as will fit on six lines.

- The ICAL run is particularly useful, since you can build a complex set of equations, one step at a time.

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- There are a few differences between ICAL and the other iterative runs. For example, a report must be on display when you execute it and the input screens are slightly different.

Ü Ü Windows Linux/UNIX OS

Iterative Run Exercise

To use iterative runs

Do this exercise to better understand the iterative runs.

1. Duplicate report 2B0 and display the duplicate report.
2. Using either the menu procedure or the control line format, start the ISOR run.
3. Enter some options and parameters. For ideas, see the *Command Reference.*
4. Press **Transmit** to see the sorted result.
5. Press **Mask** to receive the completed mask and change some parameters.
6. Press **Transmit** to receive the newly sorted result.
7. Press **Save** to display the Save Mask form.
8. Enter a mask name and press **Transmit**.
9. Press **Report** and roll to the end of the report to see the saved mask.

Printing Through Output Manager

This section includes the following topics:

- Output Manager Overview
- Printing Through Output Manager
- Output Manager Data Dependent Attribute (DDA) Designer
- Print File Overview
- Setup Overview
- Printing Examples
- Hardware and Software Requirements

Output Manager Overview

Note: The Output Manager Online Help is accessible from the "Unisys Enterprise Output Manager" Program Group.

Enterprise Output Manager software is a comprehensive output-management solution for mixed-platform networks. Running on a Windows operating system, the primary function of Output Manager software is to automatically process and route print files and other application output files from any supported platform to any supported output destination. Automated processing is based on the file name, size, or other file characteristics you specify. In addition to printing, Output Manager lets you take advantage of a variety of delivery mechanisms, such as e-mail, Web, CD/DVD, and fax. With each of these delivery methods, Output Manager can reformat text documents and data into a customized, modern appearance.

The primary benefits of Output Manager software are

- Operational cost savings
- Improved operational discipline
- Document reengineering
- Remote administration

Supported Platforms and Output

The Output Manager server software runs on IBM-compatible workstations and servers running Windows XP Professional or Windows 2003. The optional Output Manager host agent software runs on Unisys systems (Unisys ClearPath, ClearPath Plus, MCP, and OS 2200 Series systems) and manages system output from those systems. In terms of the client-server model, the Output Manager host agents and other senders of files are the "clients," and the Output Manager server software on the Windows system is the "server."

Output Manager can process output files from a variety of systems, such as Unisys ClearPath Plus and ES7000, IBM, Windows, Sun Microsystems, HP, Linux, and UNIX systems. With its standards-based open architecture, you can “unfreeze” application output files produced by various vendors, such as Oracle, PeopleSoft, and Unisys, or by customer-written application programs, such as COBOL, and other legacy applications. Output Manager can automatically customize and send these output files to the destination of your choice: a PC, file server, mainframe, e-mail system (through an e-mail client like MS Outlook or Lotus Notes), fax, CD/DVD, XML file, HTML file, printer, Web site, PDF file, customer-written program, or software application.

Output Manager accommodates both distributed and centralized print and data delivery needs. You can customize and distribute a single print file to multiple locations using any combination of delivery methods. For example, you can

- Send tailored messages and attachments to e-mail users
- Route customized invoices to a remote department’s LAN directory
- Create indexed files to be shared among users
- Send individualized facsimiles to clients
- Post the file to a Web site
- Transfer portions of a master report to various departmental printers

In summary, you can use Output Manager to distribute information directly to users, or archive destinations, from low-cost distribution points.

Printing Through Output Manager

When Output Manager detects the arrival of a BIS print file, it processes the file based upon directions contained in an Output Manager File Mask.

File Masks - Overview

Output Manager is designed to route and process files automatically. The Output Manager configuration entity that provides this function is the File Mask. A File Mask is a collection of criteria that determines how to process incoming files. File Masks are used to determine the File Group and what kind of job or jobs should be automatically created. There are dozens of items to mask on, including user-embedded information.

Examples of File Masks include

- Send large print jobs to faster printers ("If number of bytes > 20 MB, then ...")
- Identify files with special formatting requirements ("If paper type = "pretty pink", then ...")
- Identify files to convert to PDF and then e-mail the files to all Vice Presidents ("If banner-id = quarterly report, then ...")
- Distribute data according to their MCP print destination, IP address, priority, and so on

When a file arrives in Output Manager, it is compared against the File Masks, which are configured in Configuration Explorer. When the file matches the selection criteria defined in one of the File Masks, Output Manager creates a job that performs the actions specified in the File Mask.

See the Configuration Explorer topics in the Output Manager Help for more information.

Configuring a File Mask

Each File Mask has two main parts that an administrator configures through the File Masks properties pane of Configuration Explorer:

- A set of actions that the Output Manager application is to perform (such as printing, transferring, or backing up the file)
- Selection criteria that determine which files the set of actions applies to (for example, the file's creator, what host the file came from, the host queue the file came from, and the estimated pages)

In the Output Manager Help, see the following topics for additional information on the configuration procedure and configuration property values: *Configuring a File Mask* and *File Masks - Properties.*

Relationship of File Masks to BIS PR Function and PRT Statement

The value passed in the *banner* subfield of the PR function or PRT statement appears in Output Manager as the **Job Name** property of the print file. The **Job Name** property value can be displayed in the Print Job Management window of Output Manager. The value may also appear in other file properties, depending upon which BIS platform is used. Through the "Mask statements" properties of an Output Manager File Mask configuration, you can use the Job Name field to detect if the file originated from BIS.

Relationship of File Masks to File Groups

One of the actions Output Manager always performs is to assign the file to a File Group. A File Group is a set of files that are to be handled the same way after processing. The main items in the definition of a File Group are

- Whether Output Manager should maintain entries for the files in its file information database
- Whether the files should be deleted when removed from the file information database
- Where the file should be stored

If you do not specify a File Group, a default File Group is used. See *File Groups* in the Output Manager Help for more information.

Relationship of File Masks to Job Creation

Each File Mask also specifies what jobs Output Manager should create when files match the selection criteria in the File Mask.

- To create a print job, the File Mask must specify a Print Attribute and printer name.
- To create a transfer job, the File Mask must specify a Transfer Attribute and, if the destination system is an enterprise server, the name of a print queue.
- To create a backup job, the File Mask must specify a Backup Attribute and, optionally, a Print Attribute.
- To create an e-mail job, the File Mask must specify an E-mail Attribute and, optionally, a Print Attribute and/or the attachment display name.
- To create a custom job, the File Mask must specify a Custom Job Attribute and, optionally, a Print Attribute.

See *Job Types* in the Output Manager Help for more information.

Command Files

Command files specify a string of device-specific characters to be sent to the printer. This feature allows switching between emulation modes (if supported) and sending printer initiation sequences. Command files also let you send device-specific commands before printing any data. See *Command Files - Overview* in the Output Manager Help for more information.

Output Manager Data Dependent Attribute (DDA) Designer

The Output Manager Data Dependent Attribute (DDA) Designer feature is packaged separately from the standard Output Manager software and may be purchased separately to add valuable capabilities to your Output Management environment. For more information about Output Manager optional features, contact your Unisys representative.

Data Dependent Attributes - Overview

The Data Dependent Attribute (DDA) Designer, an optional feature of Output Manager, allows you to change the look of the output file without changing the application program that generated the input file. A typical case is where the legacy application is no longer supported, or is too expensive to modify. Another case is where you do not own or control the application that generates the printed output, such as a report, but you want to customize the report to fit your particular requirements.

In addition, DDA Designer lets you modernize data output, such as adding electronic forms, logos, signatures, bar codes, and more to the page – without changing the application that produced the data. Further, this allows for the centrally located Output Manager to compose the input data streams from numerous input sources, including mainframes, Linux/UNIX systems, Windows systems, and any process that can expose data to the Output Manager software.

Typically, administrators create DDAs. Once you create a DDA, you configure a Print Attribute to use it. For example, a DDA could take a simple file of invoice data that originated from a host system and reformat it for a laser printer by loading an electronic form, repositioning the data to fit on the form, and printing the postal code as a PostNet bar code.

In the Output Manager Help, see the following topics for information on creating or modifying DDAs: *DDA Designer Key* and *Configuration Explorer.*

Examples of DDA Capabilities

The following are examples of DDA capabilities:

- Move data to another location on the page
- Specify data that is not to be printed
- Change fonts for particular data
- Send command files to the printer based on data in the report
- Merge incoming data with electronic forms
- Create bars codes based on data in the report
- Dynamically switch Print Attributes based on data in the report
- Create index values for indexed files

For examples of DDA input and output, see *DDA Examples* in the Output Manager Help.

Relationship of DDAs to Print Attributes Each DDA is a sequence of line items that specify what actions to take based on the content and location of the data in a file. The DDA is invoked through a Print Attribute, which can be defined for print jobs, e-mail jobs, and custom jobs when data enters the Output Manager environment. A DDA can also be defined in a Print Attribute used to process backup jobs, but they are processed only when the data stream is composed (as when printing or viewing the file created by the Backup Job).

Unlike Print Attributes, DDA information is not kept with each print job; the print job simply references the name of

the DDA to be used by the print job. If the administrator updates a DDA, all outstanding print jobs that reference that DDA are affected. Jobs in progress while a DDA is modified are not affected.

In the Output Manager Help, see the following topics for additional information on how DDA works, DDA items, and DDA commands: *How DDA Works, More About DDA Items,* and *More About DDA Commands.*

Print File Overview

This section contains information on the following topics:

- Print File Delivery
- Executing the Print Function
- Using Print File Alternatives
- Using AUX to Route Print Files
- Delivering Print Files with the File Function

Print File Delivery

You can deliver generic print files to Output Manager using one of the following methods.

- Executing the Print Function
- Using Print File Alternatives

In most cases, you should execute the PRT statement or the PR function to route print files to Output Manager.

Executing the Print Function or PRT Statement

Use the Print (PR) function or PRT statement to print a report on a system printer. The manual function redisplays the report on your screen and places it in a print queue for the requested printer. The statement places the report in the print queue for the requested printer.

Reports printed using the PR function and the PRT statement can include special control information for Output Manager, such as e-mail addresses, text for the subject line of a mail message, and user tags. Control information is specified on the EOM printer control command. Any text beginning in column 7 of the EOM control command is appended to the “<Esc>60<Esc>S " escape sequence and written to the output print file.

You can also specify a preamble report on the PR function or PRT statement. A preamble report is a report that Output Manager processes before it processes the report specified on the PR function or the PRT statement. It can contain both fixed and dynamic control information (such as the e-mail routing information). Using a preamble report enables you to add control information without having to modify the report being printed.

The PRINTFORM script also enables you to add EOM control commands to predefined printer forms.

See the *Command Reference* for information about the PR function, the PRT statement, and the EOM printer control command. See the *Enterprise Output Manager Help* for information on user tags and the keywords you can use on escape sequences.

Using Print File Alternatives

As an alternative to executing PR or PRT to direct print files to Output Manager, you can use one of the following methods:

- Using AUX to Route Print Files
  Use the Auxiliary (AUX) function or statement to route print files to an auxiliary printer.

- Delivering Print Files with the FILE Function

You can use the Create File (FIL) statement to write the Business Information Server report to a native file and then transfer the native file to Output Manager workstation. This action transfers the file into a directory configured for the Directory Monitoring feature. When Output Manager detects the presence of a new file, it begins processing the file.

The advantage of using the Auto Directory is that it supplies additional selection criteria; the disadvantage is that you may need to change existing BIS scripts.

Using AUX to Route Print Files

You can execute the Auxiliary (AUX) function or statement to deliver print files to Output Manager. You can use LPR/LPD print queues to route auxiliary print file output to the Output Manager.

Auxiliary print to an LPR/LPD print queue is viable only with Graphical Interface for Business Information Server. Either define an LPR printer as the default for the PC, or use the Aux Printer Select command from the File selection on the menu bar to route auxiliary print to the LPR print queue. See Printing on a Windows System for specifics of defining an LPR print queue.

Existing applications that use Auxiliary (AUX) continue working with printers controlled by Output Manager. However, to realize the greatest printing flexibility, you should use Print (PR/PRT).

Delivering Print Files with the FILE Function

The FIL statement and FILE function let you direct a report to a native file.

When to Use the FILE Function

Although you should use the Print (PR/PRT) function for most printing, there are two cases where the FILE (FILE/FIL) function can provide a useful alternative for moving print data to Output Manager:

- You may need to modify the output from a script using a program external to Business Information Server before sending the file to an Output Manager based printer.

- You can use the Output Manager file name as the File Mask selection criteria. Therefore, you can create a unique handling of a class of reports in Output Manager, and identify the print file as a report belonging to that class by giving it a name which uniquely identifies it during Output Manager File Mask processing.

File Transfer Methods

You can use one of the following file transfer methods:

- If you have the Directory Monitoring feature configured, you specify a directory that Output Manager monitors for arrival of print files. When print files appear in the auto directory, Output Manager processes them

through the File Mask library. The user just needs to copy the print file to the specified directory.

- There are a few ways of transferring the file to the Output Manager workstation. FTP is the most common method for transferring files. The script could output the file with the FIL statement, and then follow it with a Linux/UNIX (Business Information Server for the Linux/UNIX Operating System) or OS (for Windows Business Information Server) command to perform the FTP from within a script.

As an alternative, you can create a CRON (Linux/UNIX System) job to periodically transfer files from a predefined local directory to the remote Output Manager workstation.

Another option is to install Business Information Server for Windows on the server that has an Output Manager workstation. You can transfer the report to that server through BIS networking (NWR), and then remotely execute the FIL statement to copy the report to the auto directory.

Using the FIL statement to drop the file into the auto directory lets you use the file name for the Output Manager File Mask selection criteria. Therefore, you can create a unique handling of a class of reports in Output Manager, and then identify the print file as a report belonging to that class. Examples include using a unique file extension (for example, *.inv), or using the first few characters of the file name (for example, inv*).

You then use the file name in the Output Manager File Mask processing to select the file for specific Output Manager processing. You can select the files by specifying the *Host File Name* to select all files with the inv extension.

Setup Overview

All of your Business Information Servers have the same print and data distribution interface, but there are system-specific requirements for each Business Information Server type to use Enterprise Output Manager. These methods assume that you configure a communication path to Enterprise Output Manager, configure communications and print queues on the host systems, and use the PRT/PR commands to send data to a print queue.

- If your Business Information Server resides on an OS 2200 host, set up the Enterprise Output Manager for ClearPath OS Component and related TSAM interface network configuration.

- If your Business Information Server resides on a Linux/UNIX Operating System host or a Windows host, define LPR/LPD print queues to enable sending data to the Output Manager through LPR/LPD protocol.

Choose a topic based on what system you are using:

- Printing on an OS 2200 System

- Printing on a Linux or UNIX System

- Printing on a Windows System

Printing on an OS 2200 System

The Business Information Server for ClearPath OS 2200 sends data through a print queue to the Enterprise Output Manager for ClearPath OS 2200 component and from there to the PC Enterprise Output Manager component on Windows.

The following presents a brief summary of the configuration steps:

1. Set up an OS 2200 print queue for Enterprise Output Manager.

   To configure a print queue for the application, use the Exec STATION statement. You must ensure that you configure at least one print queue for each Enterprise Output Manager workstation.

2. Configure the Enterprise Output Manager application in CMS 1100.

   To configure CMS 1100 for use with the Enterprise Output Manager application, the CMS 1100 configuration must include an INTERNET-ADR statement, an IP statement, and a PROCESS statement. These statements make the CMS 1100 TSAM interface available to the application.

3. If using a DCP, configure the processor as an IP router.

4. Install the Enterprise Output Manager for ClearPath OS 2200 component.

   Install the Enterprise Output Manager software on your

OS 2200 system. The Enterprise Output Manager for ClearPath OS 2200 component has only one installation mode (A).

You must set up a runstream for the Enterprise Output Manager for ClearPath OS 2200 component to tell it which queues to monitor and where to send print files from those queues. You can use any text editor to create the runstream. The runstream includes an @RUN statement, the processor call, and Output Manager configuration statements.

- Ensure security privileges are adequate.
- Ensure the Output Manager meets the quota system requirements.
- Start the Output Manager script on the host system.

Note: Refer to the *ClearPath Enterprise Servers Enterprise Output Manager for ClearPath OS 2200 and ClearPath MCP Configuration and Operations Guide* (7850 4362) for detailed documentation on installing and configuring the Enterprise Output Manager for the ClearPath OS 2200 component.

Printing on a Linux or UNIX System

Configuring the Linux or UNIX Operating System LPR Print Queue

To use LPR/LPD protocol to route Business Information Server reports to Output Manager requires defining LPR print queues. Do the following to configure these print queues.

1. Define the IP address for Output Manager workstation in the /etc/hosts file. For example, enter the following:

   `192.62.169.129` *`eom-wkstation`*

   where *eom-wkstation* is the remote system name. This action defines a network path to Output Manager.

2. Execute the lpsystem utility to register the Output Manager workstation with the local Linux or UNIX Operating System print service. This registration enables remote printing from the Linux or UNIX Operating System to *eom-wkstation*. For example, enter the following:

   `lpsystem -t bsd` *`eom-wkstation`*

3. Execute the lpadmin utility to create a local print queue and specify that files queued to this print queue are routed to *eom-wkstation*. For example, enter the following:

   `lpadmin -p` *`queue`* `-s` *`eom-wkstation`* `where` *`queue`* `is the local system printer name and` *`eom-wkstation`* `is the remote system name.`

4. Repeat the lpadmin command for any additional printers.

5. Verify this printer configuration by creating a test print, as follows:

    a. To monitor the test print, start the Output Manager application from the Windows based PC. Select Print Job Management from the task bar. The Print Job Management window appears.

    b. From the Linux or UNIX Operating System host, enter the following to create a test print:

    `lp -d` *`queue smallfile`* `where` *`queue`* `is the local system print queue and` *`smallfile`* `is the name of a test file.`

    A print preview of the test print should appear in the Print Job Management window.

Note: You must have superuser or root privileges to make these modifications.

Printing on a Windows System

Configuring the Windows System Print Queue

Make the following changes in Microsoft Windows and the Business Information Server to enable sending system print files from Microsoft Windows to the Output Manager workstation. Before starting this setup process, ensure that Microsoft TCP/IP Print Services are installed. This software is available on the Windows CD.

1. Click Start, Settings, and Printers. The Printers screen appears.

2. Click Add Printers. The Add Printer Wizard screen appears.

3. Select My Computer and click Next.

4. Click Add Port. The Printer Ports screen appears.

5. Highlight LPR Port and click New Port. The Add LPR Compatible Printer screen appears.
   If the LPR port is not an available option, the TCP/IP Print Services feature is not installed. Install this feature from the Windows Operating System CD before attempting to define an LPR print queue.

6. Enter the IP address of the system that has the Output Manager workstation installed.

7. Enter a print queue name in the field for the Name of printer or print queue that resides on the server.

   The system sends this name with the print file, and it is

accessible in Output Manager as the Host Queue display field. Give it a meaningful name because you will use this name when processing print files.

8. Click OK. The next Add Printer Wizard screen appears.

9. Select Generic as the printer type and accept the default.

10. Click Next. The next Add Printer Wizard screen appears.

11. Enter a meaningful printer name.

    Choose a name that you can use when referencing this print queue from the Business Information Server. Make a note of the name for use when configuring the system.

12. Take the defaults on the remaining dialog boxes.

13. If Output Manager is active, printing a test page places an entry in the Print Job Management window.

14. Click Finish on the final dialog box.

Configuring the Windows System Print Queue

Make the following changes in Microsoft Windows and the Business Information Server to enable sending print files from Microsoft Windows to the Output Manager workstation. Before starting this setup process, ensure that Microsoft TCP/IP Print Services are installed. This software is available on the Windows CD.

1. Click Start, Settings, and Printers. The Printers screen appears.

2. Click Add Printers. The Add Printer Wizard Screen appears.

3. Select My Computer and click Next.

4. Click Add Port. The Printer Ports screen appears.

5. Select LPR Port and click New Port. The Add LPR compatible printer screen appears.

   If the LPR port is not an available option, the TCP/IP print services feature is not installed. Install this feature from the Windows Operating System CD before attempting to define an LPR print queue.

6. Enter the IP address of the system that has the Output Manager workstation installed.

7. Enter a print queue name in the field for the Name of printer or print queue that resides on the server.

   The system sends this name with the print file, and it is

accessible in Output Manager as the Host Queue display field. Give it a meaningful name because you will use this name when processing print files.

8. Click OK. The next Add Printer Wizard screen appears.

9. Select Generic as the printer type and accept the default.

10. Click Next. The next Add Printer Wizard screen appears.

11. Enter a meaningful printer name.

    Choose a name that you can use when referencing this print queue from the Business Information Server. Make a note of the name for use when configuring the Business Information Server.

12. Take the defaults on the remaining dialog boxes.

13. If the Output Manager is active, printing a test page places an entry in the Output Manager Print Job Management window.

14. Click Finish on the final dialog box.

Directory Monitoring with Output Manager

One way to "feed" information (files) to an Output Manager server is to configure Output Manager to automatically monitor particular directories for files. These directories are called "monitored directories." For example, you can create or modify BIS print files and send them to a directory. At defined intervals, Output Manager monitors the directory for files matching your criteria. When a match is found, Output Manager processes the files through the File Mask library. Thousands of directories can be configured as monitored directories.

In the Output Manager Help, see the following topics for more information: *Configuring Output Manager to Receive Files* and *Configuring a Monitored Directory.*

Setting Up Communications Paths

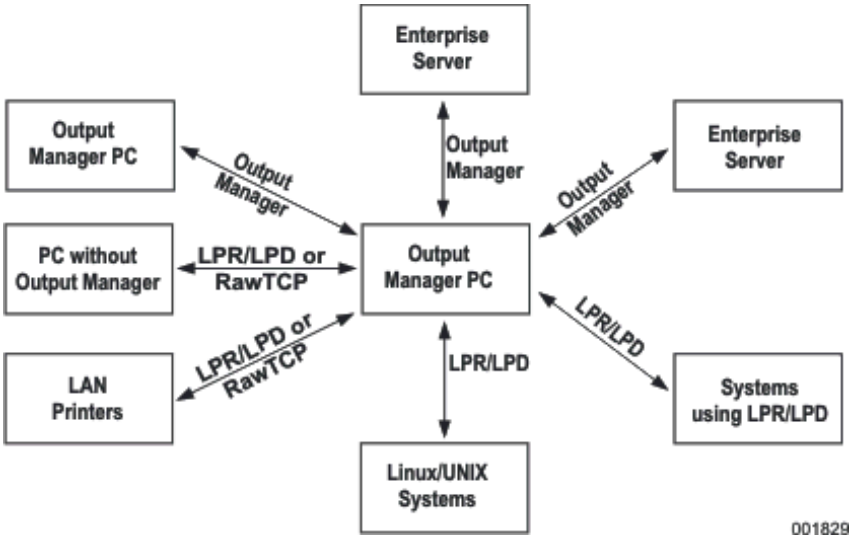
Output Manager can send files to or receive files from other Output Manager PCs/Windows servers, OS 2200 systems, LPR/LPD systems, and Raw TCP systems through TCP/IP connections referred to as communication or transfer paths.

To enable transfers, you must configure the communications paths that tell Output Manager how to reach the other system and what type of system it is. The same communications paths come into play for files moving in the other direction. The types of paths that apply to BIS are

- Communications Sending Paths
- Communications Listening Paths

The information you need to configure your communications paths depends on whether you plan to use the path to send files (called a "sending path") or to receive files (called a "listening path").

In the Output Manager Help, see the following topics for more information on configuring communications paths on the Output Manager client: *Communications Paths* and *Configuring a Communications Path.*



Additional considerations for each platform are described at [Setting Up Enterprise Output Manager to Receive Files from BIS](#).

Setting Up Enterprise Output Manager to Receive Files from BIS

To ensure that Enterprise Output Manager can receive data, you must first set up the Output Manager to receive data from Business Information Server print queues. For all platforms, the first step is to set up communication paths. See Setting Up Communications Paths for additional considerations for each platform listed below.

This topic introduces the types of systems for which you can configure Output Manager communication paths to receive files from BIS.

For details, see the topic *Sending UNIX, IBM iSeries, Linux, and Windows Print Files to Output Manager* in the Output Manager Help.

ClearPath OS 2200

The Enterprise Output Manager for ClearPath OS 2200 component is a background batch job that monitors OS 2200 print queues and sends all print files from those monitored queues to a specific Output Manager workstation. If a user executes the PRT statement or PR function to a system print queue, the Enterprise Output Manager for ClearPath OS 2200 component transfers the print file to the Output Manager workstation through a TSAM network connection.

To set up ClearPath OS 2200 Print, you must define communications paths from the Enterprise Output Manager for ClearPath OS 2200 component to the Output Manager workstation. For more information, refer to the *Enterprise*

Output Manager for ClearPath OS 2200 and ClearPath MCP Configuration and Operations Guide.

Windows or Linux/UNIX Systems

To print to Output Manager from BIS on Windows or Linux/UNIX platforms, set up an LPR/LPD printer on the BIS host machine that "points" at the Output Manager system.

LPR/LPD Systems

LPR/LPD (RFC 1179) is a standard TCP/IP protocol that originated with Berkeley UNIX. LPR/LPD is also used on other systems, including IBM mainframes with TCP/IP Services installed, iSeries systems, Sun Microsystems, HP, and Linux systems. Printers with network interfaces often receive files through the LPR/LPD protocol.

Printing Examples

This section contains the following topics:

- Printing Data Overview
- Printing a Report
- Configuring Printers and Routing Files Automatically
- Customizing Print Data

Printing Data Overview

The following examples illustrate how to print unmodified reports through Output Manager:

- Printing a Report
- Configuring Printers and Routing Files Automatically

Printing a Report

This example lists the steps performed from the Output Manager to print an unmodified Business Information Server report to a printer that is attached to the Output Manager workstation. Refer to the indicated Output Manager Help topics in Steps 1 and 2 for more information.

Step 1: Define a physical printer (see *Printer Types and Definitions* and *Configuring a Windows Printer* in the Output Manager Help)

Note: The printer must be configured in the Windows Print Manager. Then, create a Physical Printer with a Printer Type of "Windows" and the Windows Printer name property set to the name defined in the Windows Print Manager.

Step 2: Route all print files to the physical printer (see *Routing Files Automatically* in the Output Manager Help)

Step 3: Print a test report from your system

1. From the Output Manager View menu, select Printer Management. The Printer Management window is displayed.

2. On the Printer Management window, check the status of the new physical printer.

3. If the Status is locked, double-click the printer entry. The Printer Control screen appears.

4. Click Start Printing and click OK.

5. From BIS, execute a test script with the PRT statement, or execute a PR manual function that prints a report. The print queue must be configured to automatically route the print file to Output Manager.

Step 4: Monitor the printing

1. From the Output Manager View menu, select Print Job Management. The Print Job Management window appears.

2. On the Print Job Management window, you can monitor the print file arrival, watch the file as it is spooled to the specified (such as DEFAULT) printer, and observe the file printing on the physical printer.

Note: You must establish an Output Manager print queue before printing a report.

Configuring Printers and Routing Files Automatically

As was explained in Output Manager Overview, you can configure Output Manager to receive a file, such as a report, and automatically send it to a destination, such as a local or network printer.

Most of your automation and customization is achieved through the configuration settings you specify using the Configuration Explorer interface of Output Manager. In the Output Manager Help, the following topics provide procedures for accomplishing a variety of data delivery objectives: *Configuring Printers* and *Routing Files Automatically.*

Customizing Print Data

This subsection contains the following topics:

- Customizing Print Data
- Generating an Invoice from a Report

Customizing Print Data

The Output Manager capability to take a raw text file and manipulate it before printing offers the BIS user significant benefits. The following examples summarize some of the printing tasks that a user can do through Output Manager.

- Printing a BIS word processing document
- Reformatting a report with Output Manager DDA
- Generating an Invoice from a report

Refer to *Customizing Print Data* in the Output Manager Help for detailed instructions about using Output Manager to customize print data.

Generating an Invoice from a Report

The following table lists the steps you can use to generate an invoice from a report. The table also lists the Output Manager Help topic that relates to each step.

Steps to Generate an Invoice from a Report	**Enterprise Output Manager Help Topics**
Step 1: Create a BIS report/result containing the data to be processed. You must perform this step from BIS.	
Step 2: Create an electronic form	*Electronic Forms (Form Overlays)*
Step 3: Configure a DDA	*Configuring a DDA*
Step 4: Configure a print attribute	*Configuring a Print Attribute*
Step 5: Configure a File Group	*Configuring a File Group*

Step 6: Configure a File Mask	*Configuring a File Mask*
Step 7: View or print the invoice	*Viewing the File Online*

See the Output Manager Help for additional details of how to use Output Manager for customizing print data.

Hardware and Software Requirements

This section contains the following topics:

- Business Information Server for ClearPath OS 2200
- Business Information Server for Linux, UNIX Systems, and Microsoft Windows

Business Information Server for ClearPath OS 2200

The following are required for OS 2200:

- Enterprise Output Manager for ClearPath OS 2200 (included with ClearPath Systems).
- Software Library Administrator (SOLAR) or COMUS level 6R1 or higher, for installing the Enterprise Output Manager for ClearPath OS 2200 component.
- CMS 1100 level 8R2 or higher, including TSAM.
- Protocols of CMS 1100 to send and receive files from other systems over the network. The Enterprise Output Manager for ClearPath OS 2200 component uses the TSAM interface and communications.
- One of the following network connections:
    - Host LAN Controller (HLC)
    - Distributed Communications Processor (DCP)
- One or more Enterprise Output Manager workstations that meet the hardware and software requirements described in the Enterprise Output Manager documentation.

Business Information Server for Linux, UNIX Operating Systems, and Microsoft Windows

The following are required for the Linux/UNIX Operating System/Windows Client/Windows:

- Enterprise Output Manager workstation.
- An IBM compatible PC with an 80486 or higher processor.
- One of the following Windows operating systems:
  - Windows 2003
  - Windows XP
- 12-MB hard drive space for Enterprise Output Manager workstation software and sufficient space for print files.
- LPR/LPD software (this software is built into most Linux, UNIX, and Microsoft Windows operating systems).
- A CD-ROM drive and 3.5-inch diskette or access to a network for installation.
- 32-MB RAM for Windows.
- A monitor with a minimum display resolution of 800 x 600 pixels.